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PART

2

COMPLEX ANALYSIS

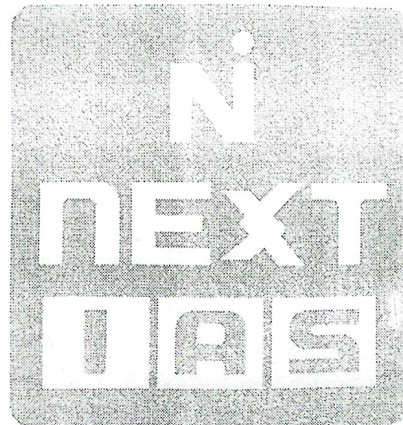
COMPLEX ANALYSIS

Basics:

- Complex number are defined as ordered pair (x, y) of real numbers and can be expressed as $x + iy$ where $x, y \in \mathbb{R}$ and i is the imaginary quantity that satisfies the equation $i^2 = -1$.

- Modulus of $Z = |z| = \sqrt{x^2 + y^2}$ and $|z|^2 = x^2 + y^2 = z\bar{z}$

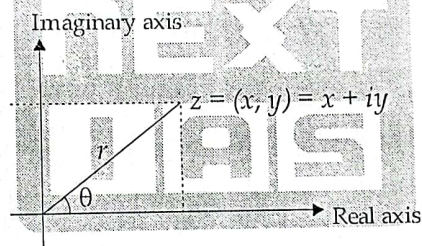
PT $|z_1 + z_2| \leq |z_1| + |z_2|$ (Triangle inequality)



Q. Prove that equality in triangle inequality holds if the two points z_1 and z_2 are collinear.

Geometric and polar representations:

Complex number: Consider them as points in Cartesian coordinate system



Polar representation:

$$z = x + iy = r(\cos \theta + i \sin \theta) \quad r, \theta \in \mathbb{R}$$

$$\Rightarrow r = \sqrt{x^2 + y^2} \text{ and } \tan \theta = y/x \text{ (where } r \text{ is non-negative and } \theta \text{ is angle that radius vector makes with +ve real axis)}$$

Note:

1. Argument of O is undefined.
2. The value $-\pi < \theta \leq \pi$ of the argument of a complex number is called principal argument and it is denoted by Arg .

$$\text{Thus } \arg z = \text{Arg } z + 2n\pi, (n = 0, \pm 1, \pm 2, \dots)$$

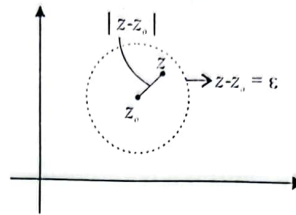
Q. Find the polar representation of $z = -\sqrt{3} - i$.



REGIONS IN THE COMPLEX PLANE

Concept of neighbourhood:

The ε -*nb*d of a point z_0 is an open disk (without its boundary) centered at z_0 and of radius ε .



It is the set of points z lying inside but not on a circle centered at z_0 (which is boundary of the disc).

Mathematically, it is given by

$$D(z_0, \varepsilon) = \{z \in \mathbb{C} : |z - z_0| < \varepsilon\}$$

Interior point of a set: A point z_0 of a set G is its interior point if there is some *nb*d of z_0 that contain only the point of G .

Exterior point of a set: z_0 is an exterior point of G if it has a *nb*d containing no point of G .

Boundary point: Neither interior nor exterior. The set of all the boundary points of G is the boundary of G e.g. the

$|z - 2|$ is the boundary of each of the sets $|z - 2| < 1$ and $|z - 2| > 1$ in the z -plane.

Open disk + boundary = closed disk $\{\bar{D} = \{z \in \mathbb{C} : |z - z_0| \leq \varepsilon\}\}$

Deleted neighbourhood of z_0 : (Open disk-center) also called punctured disk at z_0 .

e.g. $|z - 2 + i| < 1$: *nb*d of $2 - i$

$0 < |z - 2 + i| < 1$: deleted *nb*d of $2 - i$

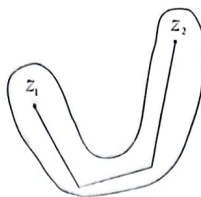
Open set: A set G of complex number is called an open set if each point of it is the centre of some ε -*nb*d which is contained in the set i.e. if z is any arbitrary point then there exist an $\varepsilon > 0$ s.t. $D(z, \varepsilon) \subseteq G$.

Note: An open set does not contain any of its boundary points.

Closed set: A set is closed if it contains all of its boundary points. For instance $|z| < 1$ is open whereas the set $|z| \leq 1$ is closed.

Note: There may be sets which are neither open nor closed and on the other hand some sets are both open and closed e.g. the punctured disk $0 < |z| \leq 1$ is neither open nor closed whereas the set of all complex numbers is both open and closed since it has no boundary points.

Open set is connected if any pair of points z_1 and z_2 in G can be joined by a polygonal lines.



An open connected set is called domain.

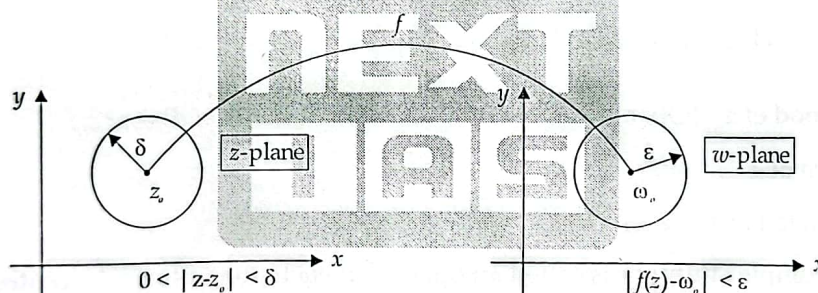
- A set G is **bounded** if there exist a real number $R > 0$ such that $|z| < R$ for every z in G .
- A point z_0 is said to be an **accumulation point** of a set G , if each deleted nbhd of z_0 contains at least one point of G .

Limits:

Suppose that a complex valued function f is defined in some nbhd $D \subseteq C$ and $z_0 \in \bar{D}$. A complex number ω_0 is said to be the **limit of function f at z_0** if for every given real number $\varepsilon > 0$, there exist a positive real number δ s.t.

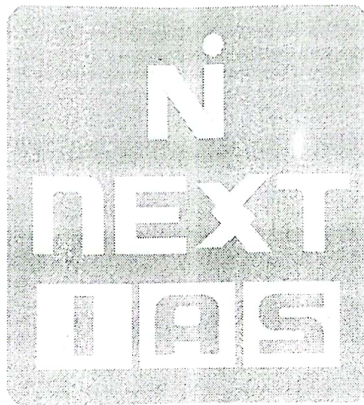
$$|f(z) - \omega_0| < \varepsilon \text{ whenever } 0 < |z - z_0| < \delta$$

Geometrically, this definition says that for every ε nbhd $|\omega - \omega_0| < \varepsilon$ of ω_0 , there is a deleted δ nbhd $0 < |z - z_0| < \delta$ of z_0 so that every z in it has its image ω_0 in the ε nbhd .



Q. Show that $\lim_{z \rightarrow 0} \frac{z^2}{z} = 0$

Q. Show that $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$ does not exist.



Continuity: A function f is said to be continuous at a point p if the following conditions are satisfied simultaneously

- (i) $f(p)$ exists
- (ii) $\lim_{z \rightarrow p} f(z)$ exist as a finite complex number
- (iii) $\lim_{z \rightarrow p} f(z) = f(p)$

or

A complex valued function f is continuous at a point p if for each positive real number ε there exists a positive real number δ s.t. $|f(z) - f(p)| < \varepsilon$ whenever z satisfies the inequality $|z - p| < \delta$.

Theorem: If $f(z) = u(x, y) + iv(x, y)$ then function f is continuous at $z_0 = x_0 + iy_0$ iff $u(x, y)$ and $v(x, y)$ are continuous at (x_0, y_0) . In other words, f is continuous if $\operatorname{Re} f$ and $\operatorname{Im} f$ are both continuous.

Note:

$$1. \quad \lim_{z \rightarrow \infty} f(z) = w \longleftrightarrow \lim_{z \rightarrow 0} f\left(\frac{1}{z}\right) = w$$

$$2. \quad \lim_{z \rightarrow z_0} f(z) = \infty \longleftrightarrow \lim_{z \rightarrow z_0} \left(\frac{1}{f(z)}\right) = 0$$

Theorem: If a function $f(z)$ is continuous and non-zero at a point p then $f(z) \neq 0$ throughout some nbd of that point.

Theorem: If a function f defined by $f(z) = u(x, y) + iv(x, y)$ is continuous throughout a closed and bounded region D , then there exist a non-negative real number M s.t. $|f(z)| \leq M$ for all points $z \in D$, where equality holds for at least one such z .



Differentiation:

A complex function $f : D \rightarrow C$, where D is an open subset of C , is said to be differentiable at $z_0 \in D$ if

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} \text{ exist. Denoted by } \frac{d}{dz} f(z_0) \text{ or } f'(z_0)$$

Note:

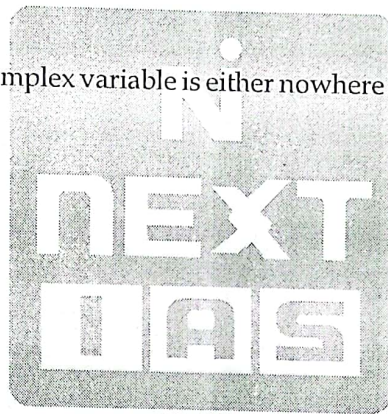
1. A function which is differentiable in the entire complex plane is called an entire function.
2. h is complex here
3. Since we cannot graph a complex function in the way we graph a real valued function.
 $f(z)$ cannot be visualised as a slope of some curves.

Theorem: If a complex valued function f is differentiable at a point z_0 , then it is continuous at z_0 .



- Q.** Show that the func $f(z) = \bar{z}$ is continuous at the point $z = 0$ but not differentiable at $z = 0$. Is this func continuous and differentiable elsewhere.

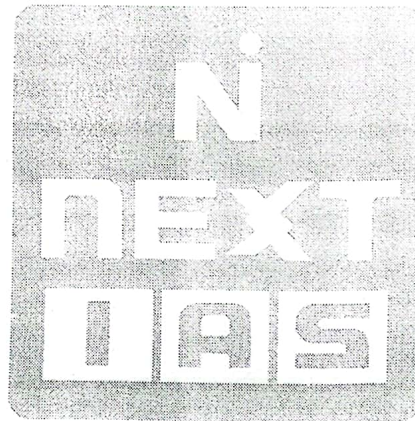
- Q. Show that real valued func of a complex variable is either nowhere differentiable or if it is differentiable then its derivative is zero.



CAUCHY REIMANN EQUATIONS

Let f be a complex differentiable function at the point $z = x + iy = (x, y)$ and $f(z) = u(x, y) + iv(x, y)$ where $u(x, y)$ and

$v(x, y)$ are real valued functions of real x and y , then $f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}$

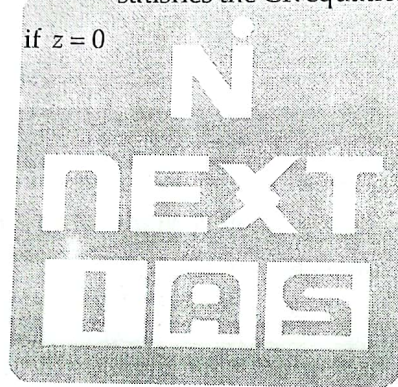


Note: CR equations are merely necessary conditions for complex differentiability of a function at a point and not the sufficient one. They are useful in locating the points where f is not differentiable (e.g. $f(z) = |z|^2$).

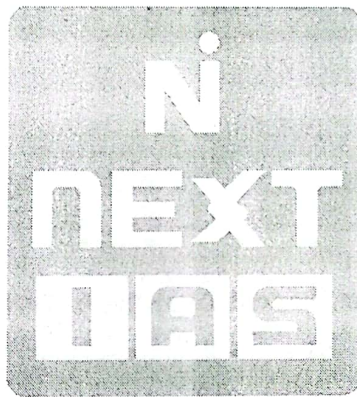
Q. Show that the function $f: \mathbb{C} \rightarrow \mathbb{C}$ given by

$$f(z) = \begin{cases} 0 & \text{if } x = 0 \\ 0 & \text{if } y = 0 \\ 1 & \text{otherwise} \end{cases} \quad \text{satisfies the CR equations at } z=0 \text{ but is not differentiable there.}$$

Q. P.T. f defined by $f(z) = \begin{cases} \frac{z^5}{|z|^4} & \text{if } z \neq 0 \\ 0 & \text{if } z = 0 \end{cases}$ satisfies the CR equation at $z = 0$ but is not differentiable there.



Polar form of CR equation:



Q. Show that $f(z) = \frac{1}{z^4}$ is differentiable at $z \neq 0$. Also obtain $f'(z)$.

Theorem: The function $f(z) = u + iv$ is differentiable at a point $z = x + iy$ if the partial derivatives u_x, u_y, v_x, v_y are continuous and satisfy the CR equation. The derivative is then given by

$$f'(z) = u_x + iv_x$$

Q. Show that $f'(z)$ exist everywhere when $f(z) = e^{-z}$ and find its value.

Analyticity:

A complex valued function f is said to be analytic (or holomorphic) at a point z_0 if f is differentiable at z_0 and at every point in some nbd of z_0 .

Notice that analyticity at a point is not the same as differentiability at that point. It is the differentiability at all points in some ε nbd of that point. Hence, being analytic is stronger than being differentiable.

Q. Show that the function $f(z) = |z|^2$ is differentiable at $z = 0$, but is nowhere analytic.

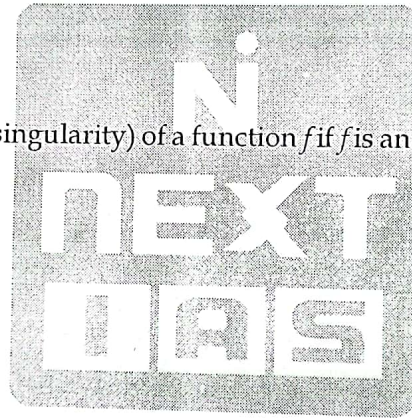


Q. Show that the function $f(z) = x^2 + y^2 + i2xy$ is nowhere analytic.

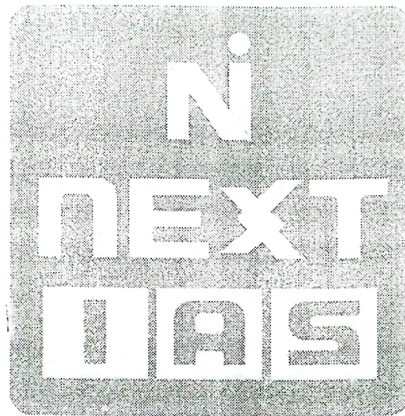
Singular point:

A point z_0 is called a singular point (or singularity) of a function f if f is analytic at some point in every nbd of z_0 but fails to be analytic at z_0 .

Consider $f(z) = 1/z$ and $f(z) = |z|^2$.



Note: Suppose $u(x, y)$ has partial derivative u_x and u_y that vanish at every point of the domain D , then u is a constant throughout D .

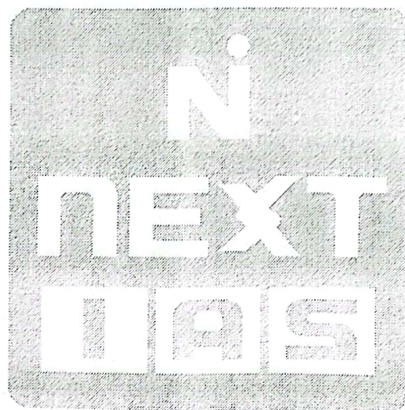


Now consider the following proposition:

Suppose f is an analytic function in domain D , then f will be a constant function if any of the following conditions are satisfied:

1. $f(z) = 0 \quad (\forall z \in D)$
2. $|f(z)|$ is const in D
3. $f(z)$ is real $\forall z \in D$
4. $\arg f(z)$ is const in D
5. $\operatorname{Re} f$ is const in D
6. $\operatorname{Im} f$ is const in D

Note: A function f such that $\bar{f}$ is analytic is called anti analytic.



HARMONIC FUNCTIONS

A real valued function N of two real variables defined on a domain D of xy -plane i.e.

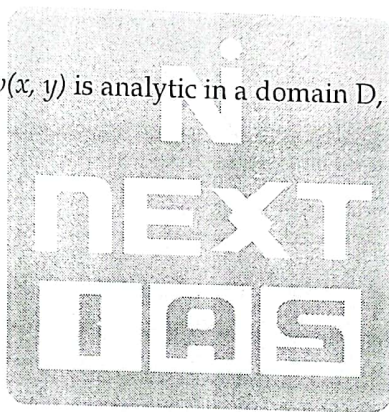
$H: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is called harmonic function if the first and second order partial derivations of H w.r.t variables x and y exist and are continuous everywhere in D and there second order partial derivatives satisfy the differential equation

$\nabla^2 H \equiv \frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} = 0, (x, y) \in D$ known as Laplace's equation. Hence, ∇^2 is the second order differential operator called the Laplacian.

Q. Show that the functions $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$ are harmonic.



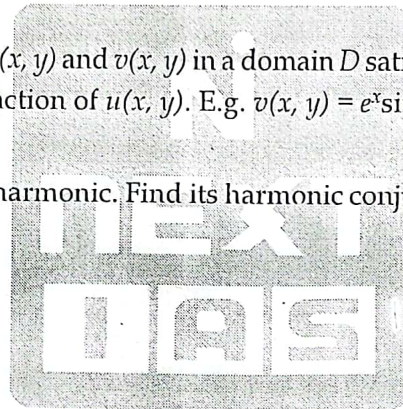
Theorem: If a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then its component function u and v are harmonic in D .

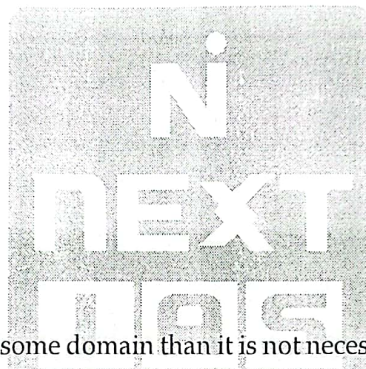


Note: We cannot choose two arbitrary harmonic functions u and v and claim that the resulting function $f = u + iv$ is analytic. e.g. $u(x, y) = x$ and $v(x, y) = -y$ are harmonic function in C but $f = u + iv = x - iy$ is nowhere analytic. Infact it is nowhere differentiable.

Definition: If two harmonic functions $u(x, y)$ and $v(x, y)$ in a domain D satisfy the C-R equations through D then $v(x, y)$ is called the harmonic conjugate function of $u(x, y)$. E.g. $v(x, y) = e^x \sin y$ is the harmonic conjugate of $u(x, y) = e^x \cos y$.

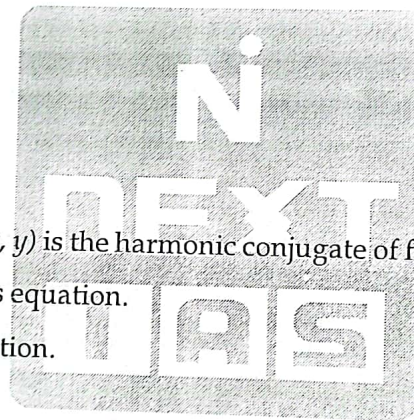
Q. If the function $u(x, y) = x^2 - y^2$ is harmonic. Find its harmonic conjugate.





Note: If v is harmonic conjugate of u in some domain then it is not necessary that u is a harmonic conjugate of v .

Q. Find the conjugate harmonic function $v(x, y)$ corresponding to $u(x, y) = x^2 - y^2 + x$ on the domain $|z| < \infty$.



Summary: If a real valued function $v(x, y)$ is the harmonic conjugate of func $u(x, y)$ in a given domain, then

- (i) both u and v satisfy the Laplace's equation.
- (ii) both u and v satisfy the C-R equation.
- (iii) $f(z)$ is analytic in the domain.

METHODS OF CONSTRUCTION A REGULAR FUNCTION:

(i) Milne-Thomson's method:

Let $f(z) = u(x, y) + iv(x, y)$ and $u_x = \phi_1(x, y); u_y = \phi_2(x, y)$

Then,
$$f(z) = \int \phi_1(z, 0) dz - i \int \phi_2(z, 0) dz + C$$

Q. If $u = e^x(x \cos y - y \sin y)$, find the analytic function $u + iv$.



Method II: If the real part of an analytic function $f(z)$ is a given harmonic function $u(x, y)$.

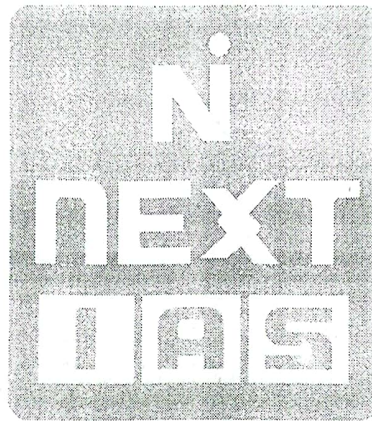
$$f(z) = 2u(z/2, z/2i) - u(0, 0) + Ci, \text{ where } C \text{ is real}$$

Q. Construct the analytic func $f(z)$, where $u = x^3 - 3xy^2 + 3x + 1$



COMPLEX INTEGRATION

- Q. Prove that the value of the integral of $1/z$ along a semi-circular arc $|z| = 1$ from -1 to 1 is $-\pi i$ or πi according as the arc lies above or below the real axis.



Theorem: If a function $f(z)$ is continuous on a contour L of length l and if M be the upper bound of $|f(z)|$ on L i.e.

$$|f(z)| \leq M \text{ on } L, \text{ then } \int_L f(z) dz \leq ml$$

Note: The integral $\int f(z) dz$ with continuous f , depends only on the end points of L iff f is the derivative of an analytic function in D .

Cauchy's fundamental theorem:

If $f(z)$ is analytic at the points within and on the closed contour C , then $\int_C f(z) dz = 0$.

Cauchy Goursat theorem:

In Cauchy's theorem, the function $f'(z)$ is assumed to be a continuous function. It was Goursat who proved that the theorem is valid without considering the continuity of $f'(z)$.

i.e. Let $f(z)$ be analytic in a simply connected domain D and let C be any closed continuous rectifiable curve in D ,

$$\text{then } \int_C f(z) dz = 0$$

Corollary: Let D be a doubly connected region bounded by two simple closed curves C_1 and C_2 such that C_2 is contained in C_1 and $f(z)$ is analytic in the region between these curves and continuous on C_1 , then

$$\int_{C_1} f(z) dz = \int_{C_2} f(z) dz \text{ where both } C_1 \text{ and } C_2 \text{ are traversed in positive sense.}$$

Cauchy's integral formula:

Let $f(z)$ be an analytic function in a simple connected domain D enclosed by rectifiable Jordan curve C and let $f(z)$

be continuous on C , then $f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz$ where z_0 is any point of D .

Gauss's mean value theorem:

If $f(z)$ is an analytic function on a domain D and if the circular region $|z - z_0| \leq \rho$ is contained in D , then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{i\theta}) d\theta$$

i.e. the value of $f(z)$ at the point z_0 equals the average of its values on the boundary of the circle $|z - z_0| = \rho$

Derivative of analytic function:

1. Let $f(z)$ be analytic function within and on the boundary C of a simple connected region D and let z_0 be any point with C , then

$$f'(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^2} dz$$

$$2. \quad f^n(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

Cauchy's inequality:

Let $f(z)$ be analytic in a domain D and let D contain the interior and the boundary of the circle γ defined by $|z - z_0| = \rho$ and if $|f(z)| \leq M$ on γ , then $|f^n(z_0)| \leq n! \frac{M}{\rho^n}$

Example: If $f(z) = \frac{z^2 + 5z + 6}{z - 2}$, does Cauchy's theorem apply

- (i) When the path of integration C is a circle of radius 3 with origin as centre.
 (ii) When C is a circle of radius 1 with origin as centre.

Q. Evaluate $\int_C \frac{1}{z(z-1)} dz$ where C is the circle $|z| = 3$.

Note:

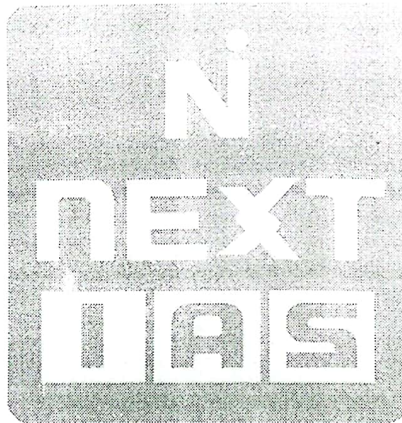
1. If $f(z)$ is a single valued analytic function in a simply connected domain, then for $a, b \in D$ we have

$$\int_a^b f(z) dz = F(b) - F(a), \text{ where } F(z) \text{ is any indefinite integral of } f(z).$$

2. Integral function or entire function - analytic in every finite region of z -plane.

Liouville's theorem:

If $f(z)$ is an integral function satisfying the inequality $|f(z)| \leq M \forall z$, where M is a positive constant, then $f(z)$ is constant.



EXPANSION OF ANALYTIC FUNCTION AS POWER SERIES

Taylor's theorem:

Let $f(z)$ be analytic at all points within a circle C_0 with centre z_0 and radius r_0 , then for every point with C_0 , we have

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \dots + \frac{f^{(n)}(z_0)}{n!}(z - z_0)^n + \dots$$

$$= f(z_0) + \sum_{n=1}^{\infty} \frac{(z - z_0)^n}{n!} f^{(n)}(z_0)$$

Laurent's theorem:

Let $f(z)$ be analytic in the ring shaped region D bounded by two concentric circles C_1 and C_2 with centre z_0 and radii r_1 and r_2 ($r_1 > r_2$) and let z be any point of D , then

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - z_0)^n}$$

where, $a_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(w)}{(w - z_0)^{n+1}} dw$ and $b_n = \frac{1}{2\pi i} \int_{C_2} \frac{f(w)}{(w - z_0)^{-n+1}} dw$

Q. Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent's series valid for the regions

(i) $|z| < 1$

(ii) $1 < |z| < 3$

(iii) $|z| > 3$

(iv) $0 < |z+1| < 2$

Q. Find the Laurent expansion of $\frac{z}{(z+1)(z+2)}$ about the singularity $z = -2$. Specify the region of convergence.



SINGULARITIES OF AN ANALYTIC FUNCTION

If a function is analytic at all points of a bounded domain except at a finite number of points then these exceptional points are called singular points or singularities.

Isolated and non-isolated singularities:

If z_0 is a singularity of $f(z)$ and if $f(z)$ is analytic at each point in some neighbourhood of z_0 , then z_0 is called an isolated singularity of $f(z)$ otherwise it is called non-isolated singularity.

e.g.

$$1. \quad f(z) = \frac{z^2 + 5}{z(z-3)(z^2+1)}$$

$$2. \quad \log z$$

The zeros of an analytic function:

The value of z for which the analytic function $f(z)$ becomes zero is said to be the zero of $f(z)$.

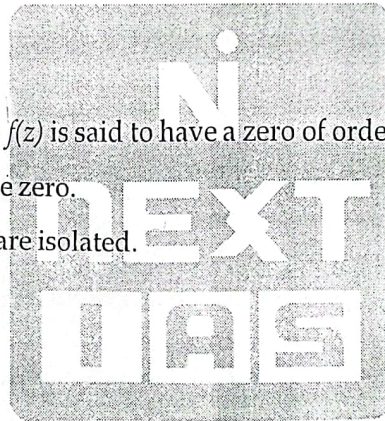
As $f(z)$ is analytic, we can expand $f(z)$ as Taylor's series about $z = z_0$.

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

If $a_0 = a_1 = a_2 = \dots = a_{m-1} = 0$ and $a_m \neq 0$, $f(z)$ is said to have a zero of order m at $z = z_0$.

A zero of order one is said to be a simple zero.

Note: The zeros of an analytic function are isolated.



Now, consider $f(z)$, analytic in a domain D except at the point $z = z_0$, then there exists a deleted neighbourhood $0 < |z - z_0| < R$ in which $f(z)$ is analytic. In the annulus $0 < |z - z_0| < R$ the Laurent expansion of $f(z)$ is

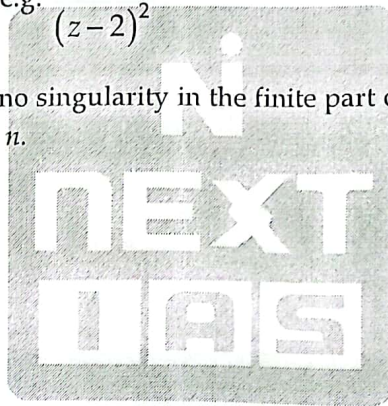
$$f(z) = \sum a_n (z - z_0)^n + \underbrace{\sum b_n (z - z_0)^{-n}}_{\text{(this part is principal part of } f(z) \text{ at } z = z_0)}$$

(i) Principal part has infinite terms $\rightarrow z_0$ is isolated essential singularity e.g. $\sin 1/z$

(ii) No term in principal part $\rightarrow$ removable singularity at z_0 e.g. $\frac{\sin(z-1)}{(z-1)}$

(iii) Finite terms in principal part $\rightarrow$ e.g. $\frac{z^2}{(z-2)^2}$

Q. Show that a function which has no singularity in the finite part of the plane and has a pole of order n at infinity is a polynomial of degree n .



- Q. If a function $f(z)$ is analytic for all finite values of z and as $|z| \rightarrow \infty$, $|f(z)| = a|z|^k$, then $f(z)$ is a polynomial of degree $\leq k$.

Theorem on poles and other singularities:

1. A function which has no singularity in the finite part of the plane or at infinity is constant.
2. Poles are isolated.

Theorem: If $f(z)$ is a function such that for some positive integer m , a value $\phi(z_0)$ exists with $\phi(z_0) \neq 0$ s.t. the function $\phi(z) = (z - z_0)^m f(z)$ is analytic at z_0 , then $f(z)$ has pole of order m at z_0 .



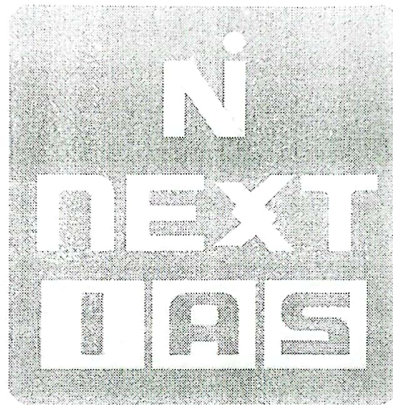
Limit points of zeros:

1. Let $f(z)$ be analytic in a domain D and let E be a set of zeros of $f(z)$ having a limit point α in D . Then $f(z)$ vanishes for all $z \in D$

2. If $f(z)$ and $g(z)$ are analytic in a domain D and if $f(z) = g(z)$ on a subset E of D which has a limit point α in D , then $f(z) = g(z)$ in the whole of D .

Limit points of poles:

1. The limit point of a sequence of poles of a function $f(z)$ is non-isolated essential singularity.



Maximum modulus principle:

Let $f(z)$ be analytic within and on a simple closed contour C . The $|f(z)|$ reaches its maximum value on C , unless $f(z)$ is constant.



Minimum modulus principle:

If m is the minimum of $|f(z)|$ inside and on C , then unless f is constant, $|f(z)| > m$ for every point z inside C i.e. $|f(z)|$ reaches its minimum value on C .

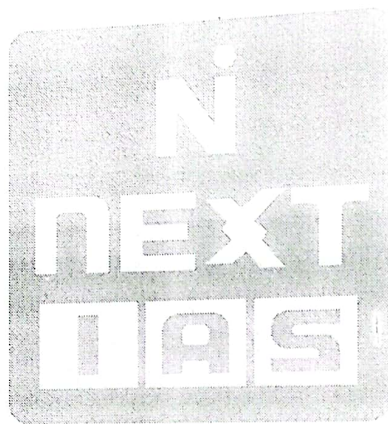
Excess of zeroes over poles:

If $f(z)$ is meromorphic inside a closed contour C and has no zero on C , then $\frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} dz = N - P$ where N is the number of zeroes and P is the number of poles inside C , (a pole or zero of order m must be counted m times).

Rouche's theorem:

Let $f(z)$ and $g(z)$ be analytic inside and on a simple closed curve C and let $|g(z)| < |f(z)|$ on C . Then $f(z)$ and $f(z) + g(z)$ have the same number of zeros inside C .

Q. Prove that all the roots of $z^7 - 5z^3 + 12 = 0$ lie between the circle $|z| = 1$ and $|z| = 2$.



Residue at a Pole

Definition: Let $z = a$ be a pole of order m of a single-valued function $f(z)$ and γ be any circle of radius r and centre $z = a$ containing no other singularities except $z = a$. Then the function $f(z)$ is regular within the region $0 < |z - a| < r$ so we can expand $f(z)$ in a Laurent's series in the region $0 < |z - a| < r$.

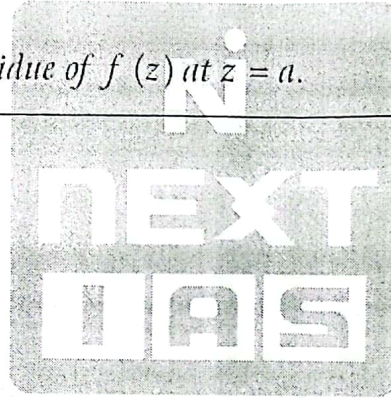
$$\therefore f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$

where
$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{(z-a)^{n+1}} dz$$

and
$$b_n = \frac{1}{2\pi i} \int_{\gamma} (z-a)^{n-1} f(z) dz.$$

In particular,
$$b_1 = \frac{1}{2\pi i} \int_{\gamma} f(z) dz.$$

The coefficient b_1 is called the residue of $f(z)$ at $z = a$.



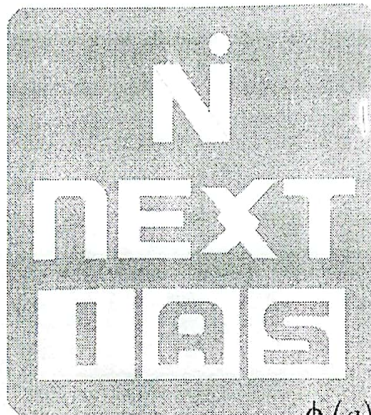
Computation of Residue at a Finite Pole

(i) Residue at a simple pole: If $z = a$ is a simple pole of $f(z)$ then the Laurent's expansion of $f(z)$ is of the form

Hence the residue at the simple pole $z = a$ is given by

$$\lim_{z \rightarrow a} (z - a) f(z)$$

Another form is obtained as follows :



Hence the residue at the simple pole $z = a$ is $\frac{\phi(a)}{\psi'(a)}$.

(ii) Residue at a pole of order greater than unity :
(Residue at a pole of order m)

$$M_1 = \frac{\phi^{(m-1)}(a)}{(m-1)!} \text{ which is the required residue at } z = a.$$

In particular if $f(z) = \frac{\phi(z)}{(z-a)^2}$, the residue at $z = a$ is $\phi'(a)$ and if $f(z) = \frac{\phi(z)}{(z-a)^3}$, the residue at $z = a$ is $\frac{\phi''(a)}{2!}$ etc.

Alternative Proof (a): Suppose $z = a$ is a pole of order m . Then $f(z)$ is of the form $\frac{\phi(z)}{(z-a)^m}$ where $\phi(z)$ is analytic.

Residue of $f(z)$ at $z = a$ is given by

$$\begin{aligned} b_1 &= \frac{1}{2\pi i} \int_{\gamma} f(z) dz = \frac{1}{2\pi i} \int_{\gamma} \frac{\phi(z)}{(z-a)^m} dz \\ &= \frac{\phi^{(m-1)}(a)}{(m-1)!}, \text{ by Cauchy's integral formula.} \end{aligned}$$

Hence the residue of $f(z)$ at the pole of order m is $\frac{\phi^{(m-1)}(a)}{(m-1)!}$, where $z = a$ is the pole of order m .

Alternative Proof (b): If $z = a$ is the pole of order m of $f(z)$ then we have

$$f(z) = \frac{\phi(z)}{(z-a)^m} \text{ where } \phi(z) \text{ is analytic at } z = a.$$

The residue at $z = a$ is the coefficient of $(z-a)^{-1}$ in $f(z)$

$$\begin{aligned} &= \text{coeff. of } (z-a)^{m-1} \text{ in } \phi(z) \\ &= \text{coeff. of } (z-a)^{m-1} \text{ in } [\phi(a) + (z-a)\phi'(a) + \dots \\ &\quad + \frac{(z-a)^{m-1}}{(m-1)!} \phi^{(m-1)}(a) + \dots] \\ &= \frac{\phi^{(m-1)}(a)}{(m-1)!}. \end{aligned}$$

Note: to find the residue of $f(z)$ at $z = a$, put $z = a + t$ in $f(z)$ and expand in powers of t , where t is small, the coefficient of $1/t$ is the residue at $z = a$.

Residue at Infinity

Definition: If the function $f(z)$ has an isolated singularity at infinity or is analytic there then the residue of $f(z)$ at $z = \infty$ is given by

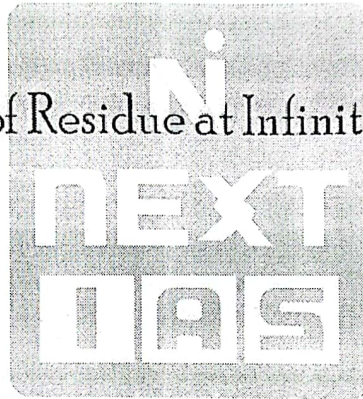
$$\frac{1}{2\pi i} \int_C f(z) dz$$

where C is a large circle containing all the finite singularities of $f(z)$ and integral along C is performed in a clockwise direction provided that this integral has a definite value.

If the integral along C is taken in anti-clockwise direction the residue at infinity is

$$-\frac{1}{2\pi i} \int_C f(z) dz.$$

Computation of Residue at Infinity



An Important Observation: We can easily show that the residue of a function at a finite point is zero if the function is analytic there. On the other hand a function may be analytic at $z = \infty$ but yet has a residue there.

Consider $f(z) = \frac{1}{z-a}$. The only singularity of $f(z)$ is a simple pole at $z = a$. The function is analytic at $z = \infty$.

$$\begin{aligned} \text{Now the residue at infinity} &= -\frac{1}{2\pi i} \int_C \frac{dz}{z-a} \\ &= -\frac{1}{2\pi i} \int_0^{2\pi} \frac{re^{i\theta}}{re^{i\theta}} d\theta, \text{ putting } z-a = re^{i\theta} \\ &= -\frac{1}{2\pi} \int_0^{2\pi} d\theta = -1. \end{aligned}$$

Cauchy's Residue Theorem

If $f(z)$ is regular, except at a finite number of poles within a closed contour C and continuous on the boundary of C , then

$$\int_C f(z) dz = 2\pi i \Sigma R,$$

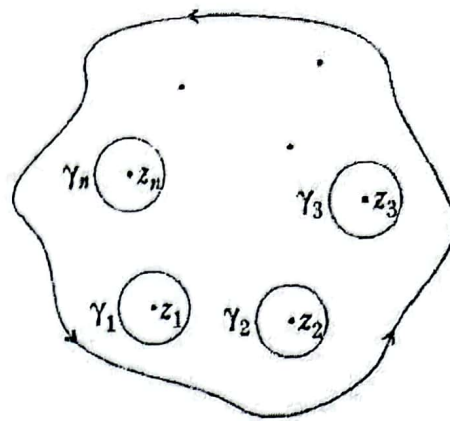
where ΣR is the sum of the residues of $f(z)$ at its poles within C .

Proof: Let $z_1, z_2, \dots, z_n$ be the n poles within the closed contour C . Let $\gamma_1, \gamma_2, \dots, \gamma_n$ be the circles with centres $z_1, z_2, \dots, z_n$ respectively and each of radius r so small that all the circles lie entirely within C and do not overlap. Then $f(z)$ is analytic in the region lying between C and the circles. Then by Cauchy's theorem

$$\int_C f(z) dz - \sum_{k=1}^n \int_{\gamma_k} f(z) dz = 0$$

or

$$\int_C f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz + \dots + \int_{\gamma_n} f(z) dz. \quad \dots(1)$$



Suppose $f(z)$ has a pole of order m_1 at $z = z_1$ then we have

$$f(z) = \phi_1(z) + \sum_{k=1}^{m_1} \frac{b_k}{(z - z_1)^k}, \text{ where } \phi_1 \text{ is analytic within and on } \gamma_1.$$

We have $\int_{\gamma_1} \phi_1(z) dz = 0$ since ϕ_1 is analytic within and on γ_1 .

Also

$$\int_{\gamma_1} \frac{b_1}{z - z_1} dz = \int_0^{2\pi} \frac{b_1 r i e^{i\theta}}{r e^{i\theta}} d\theta, \text{ putting } z - z_1 = r e^{i\theta}$$

$$= \int_0^{2\pi} b_1 i d\theta = 2\pi i b_1$$

and

$$\int_{\gamma_1} \frac{b_{m_1}}{(z - z_1)^{m_1}} dz = \int_0^{2\pi} \frac{b_{m_1} r i e^{i\theta}}{r^{m_1} e^{i m_1 \theta}} d\theta, \text{ putting } z - z_1 = r e^{i\theta}$$

$$= \frac{i b_{m_1}}{r^{m_1-1}} \int_0^{2\pi} e^{-i(m_1-1)\theta} d\theta = 0, m_1 \neq 1.$$

Substituting these values in (2), we get

$$\int_{\gamma_1} f(z) dz = 2\pi i b_1 = 2\pi i \times \text{residue of } f(z) \text{ at } z = z_1.$$

Proceeding as above, we have

$$\int_{\gamma_2} f(z) dz = 2\pi i \times \text{residue of } f(z) \text{ at } z = z_2 \text{ and so on.}$$

Hence from (1), we have

$$\int_C f(z) dz = 2\pi i (\text{sum of the residues at } z_1, z_2, \dots, z_n)$$

$$= 2\pi i \sum R.$$

Corollary: If an analytic function has singularities at a finite number of points (including that at infinity), then the sum of the residues at these points along with infinity is zero.

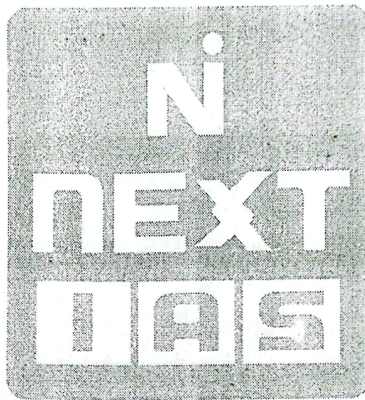
Let C be the circle enclosing within it all the singularities excluding infinity. Then by the previous theorem, we have

$\frac{1}{2\pi i} \int_C f(z) dz =$ sum of the residues at all the finite singular points within C ,

also the residue at infinity is $-\frac{1}{2\pi i} \int_C f(z) dz$.

Hence the sum of the residues at all the finite poles along with infinity is zero.

Find the residue of $\frac{z^3}{(z-1)^4(z-2)(z-3)}$ at $z=1$.



Determine the poles of the function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$

and the residue at each point.

Evaluation of Real Definite Integrals by Contour Integration

Theorem 1: If $\lim_{z \rightarrow a} (z - a) f(z) = A$ and if C is the arc $\theta_1 \leq \theta \leq \theta_2$ of the circle $|z - a| = r$, then $\lim_{r \rightarrow 0} \int_C f(z) dz = i A (\theta_2 - \theta_1)$.

In particular, if $(z - a) f(z) \rightarrow 0$ as $z \rightarrow 0$, then we have

$$\int_C f(z) dz \rightarrow 0 \text{ as } z \rightarrow 0.$$

Theorem 2: If C is an arc $\theta_1 \leq \theta \leq \theta_2$ of the circle $|z| = R$ and if $\lim_{R \rightarrow \infty} z f(z) = A$ then

$$\lim_{R \rightarrow \infty} \int_C f(z) dz = i(\theta_2 - \theta_1) A.$$

Integration Round the Unit Circle

We shall discuss here the method of evaluation, by contour integration, of the integrals which are of the type

$$\int_0^{2\pi} f(\cos \theta, \sin \theta) d\theta$$

where integrand is a rational function of $\cos \theta$ and $\sin \theta$.

Putting $z = e^{i\theta}$, we have $dz = i e^{i\theta} d\theta$

and $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$ and $\sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right).$

Now
$$\int_0^{2\pi} f(\cos \theta, \sin \theta) d\theta = \frac{1}{i} \int_C f \left[\frac{1}{2} \left(z + \frac{1}{z} \right), \frac{1}{2i} \left(z - \frac{1}{z} \right) \right] \frac{dz}{z}$$

$$= \int_C \phi(z) dz, \text{ say,}$$

where C is the circle $|z| = 1$.

It is obvious that $F(z)$ is a rational function of z .

Thus by residue theorem we have

$$\int_C \phi(z) dz = 2\pi i \Sigma R_C;$$

where ΣR_C is the sum of the residues of $\phi(z)$ at its poles inside C .

Q.

Show that

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \int_0^{2\pi} \frac{d\theta}{a + b \sin \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}, \quad a > b > 0.$$

$$= \int_0^{2\pi} \frac{d\theta}{a + \frac{1}{2}b(e^{i\theta} + e^{-i\theta})} = \frac{1}{i} \int_C \frac{dz}{z \left\{ a + \frac{1}{2}b \left(z + \frac{1}{z} \right) \right\}},$$

putting $e^{i\theta} = z, i e^{i\theta} d\theta = dz$

$$= \frac{2}{i} \int_C \frac{dz}{bz^2 + 2az + b}, \text{ where } C \text{ is the unit circle } |z| = 1$$

The poles of the integrand $f(z) = \frac{2}{i(bz^2 + 2az + b)}$ are given by

$$bz^2 + 2az + b = 0$$

or
$$z = \frac{-2a \pm \sqrt{(2a)^2 - 4b^2}}{2b} = \frac{-a \pm \sqrt{a^2 - b^2}}{b}.$$

Let
$$\alpha = \frac{-a + \sqrt{a^2 - b^2}}{b} \quad \text{and} \quad \beta = \frac{-a - \sqrt{a^2 - b^2}}{b}.$$

Since $a > b > 0$ therefore $|\beta| > 1$. Also $|\alpha\beta| = 1$ so we have $|\alpha| < 1$.

Thus $z = \alpha$ is the simple pole lying inside C .

Since α, β are the roots of $bz^2 + 2az + b = 0$, therefore we have

$$f(z) = \frac{2}{ib(z - \alpha)(z - \beta)}.$$

Residue of $f(z)$ at the simple pole $z = \alpha$ is

$$\begin{aligned} &= \lim_{z \rightarrow \alpha} (z - \alpha) \frac{2}{ib(z - \alpha)(z - \beta)} = \lim_{z \rightarrow \alpha} \frac{2}{b i (z - \beta)} \\ &= \frac{2}{b i (\alpha - \beta)} = \frac{2}{b i \frac{2\sqrt{a^2 - b^2}}{b}} = \frac{1}{i \sqrt{a^2 - b^2}}. \end{aligned}$$

Hence, $\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = 2\pi i$ sum of the residues of $f(z)$ at the poles inside C

$$= 2\pi i \cdot \frac{1}{i \sqrt{a^2 - b^2}} = \frac{2\pi}{\sqrt{a^2 - b^2}}.$$

Similarly, we can evaluate $\int_0^{2\pi} \frac{d\theta}{a + b \sin \theta}.$

Q. Evaluate $\int_{-\pi}^{2\pi} \frac{a \cos \theta}{a + \cos \theta} d\theta, a > 1$

by putting $e^{i\theta} = z$ and using the theory of residues,

Solution: Let
$$I = \int_{-\pi}^{\pi} \frac{a \cos \theta}{a + \cos \theta} d\theta = 2 \int_0^{\pi} \frac{a \cos \theta}{a + \cos \theta} d\theta$$

$$= \int_0^{2\pi} \frac{2a \cos \theta}{2a + 2 \cos \theta} d\theta$$

$$\begin{aligned}
 &= \text{real part of } \int_0^{2\pi} \frac{2a e^{i\theta}}{2a + 2 \cos \theta} d\theta \\
 &= \text{real part of } \int_C \frac{2az}{2a + z + \frac{1}{z}} \cdot \frac{dz}{iz} \\
 &= \text{real part of } \int_C \frac{2az}{i(z^2 + 2az + 1)} dz \\
 &= \text{real part of } \int_C f(z) dz, \text{ say,} \quad \dots(1)
 \end{aligned}$$

where C is the unit circle $|z| = 1$

Solving $z^2 + 2az + 1 = 0$, we get

$$z = -a + \sqrt{a^2 - 1} = \alpha, \quad z = -a - \sqrt{a^2 - 1} = \beta,$$

which are the simple poles of $f(z)$.

Since $|\beta| > 1$ and $|\alpha\beta| = 1$ therefore $|\alpha| < 1$. Thus $z = \alpha$ is the simple pole inside C .

$$\begin{aligned}
 \text{Residue at } z = \alpha &\text{ is } \lim_{z \rightarrow \alpha} (z - \alpha) f(z) \\
 &= \lim_{z \rightarrow \alpha} \frac{(z - \alpha) 2az}{i(z - \alpha)(z - \beta)} = \lim_{z \rightarrow \alpha} \frac{2az}{i(z - \beta)} = \frac{2a\alpha}{i(\alpha - \beta)} \\
 &= \frac{2a \{-a + \sqrt{a^2 - 1}\}}{2i \sqrt{a^2 - 1}} = ai \left\{ \frac{a}{\sqrt{a^2 - 1}} - 1 \right\}.
 \end{aligned}$$

$\therefore$ by Cauchy's residue theorem, we have

$$\int_C f(z) dz = 2\pi i \cdot ai \left\{ \frac{a}{\sqrt{a^2 - 1}} - 1 \right\} = 2a\pi \left\{ 1 - \frac{a}{\sqrt{a^2 - 1}} \right\}.$$

$$\begin{aligned}
 \text{Hence, } \int_{-\pi}^{\pi} \frac{a \cos \theta}{a + \cos \theta} d\theta &= \text{real part of } \int_C f(z) dz, \text{ from (1)} \\
 &= 2a\pi \left\{ 1 - \frac{a}{\sqrt{a^2 - 1}} \right\}.
 \end{aligned}$$

Q Prove that $\int_0^{2\pi} \frac{\cos^2 3\theta}{1 - 2p \cos 2\theta + p^2} d\theta = \pi \frac{1 - p + p^2}{1 - p}, 0 < p < 1.$

Solution: We have $I = \int_0^{2\pi} \frac{\cos^2 3\theta}{1 - 2p \cos 2\theta + p^2} d\theta$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^{2\pi} \frac{1 + \cos 6\theta}{1 - 2p \cos 2\theta + p^2} d\theta \\
 &= \frac{1}{2} \text{ real part of } \int_0^{2\pi} \frac{1 + e^{i6\theta}}{1 - 2p \cos 2\theta + p^2} d\theta
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \text{ real part of } \int_C \frac{1+z^6}{1-p\left(z^2+\frac{1}{z^2}\right)+p^2} \cdot \frac{dz}{iz}, \text{ putting } z = e^{i\theta} \\
&= \frac{1}{2} \text{ real part of } \int_C \frac{z(1+z^6)}{i(1-pz^2)(z^2-p)} dz \\
&= \frac{1}{2} \text{ real part of } \int_C f(z) dz, \text{ say,}
\end{aligned}$$

where C is the unit circle.

$z = \pm \sqrt{p}, \pm 1/\sqrt{p}$ are the simple poles of $f(z)$.

Since $0 < p < 1$ therefore $z = \sqrt{p}, -\sqrt{p}$ are the only simple poles which lie within C .

Residue at $z = \sqrt{p}$ is

$$\lim_{z \rightarrow \sqrt{p}} (z - \sqrt{p}) f(z) = \lim_{z \rightarrow \sqrt{p}} \frac{(z - \sqrt{p}) z (1+z^6)}{i(1-pz^2)(z^2-p)} = \frac{1}{2i} \left(\frac{1+p^3}{1-p^2} \right)$$

and residue at $z = -\sqrt{p}$ is

$$\lim_{z \rightarrow -\sqrt{p}} (z + \sqrt{p}) f(z) = \lim_{z \rightarrow -\sqrt{p}} (z + \sqrt{p}) \frac{z(1+z^6)}{i(1-pz^2)(z^2-p)} = \frac{1}{2i} \left(\frac{1+p^3}{1-p^2} \right)$$

$\therefore$ Sum of the residues at the poles inside C

$$= \frac{1}{i} \left(\frac{1+p^3}{1-p^2} \right).$$

Hence,
$$\int_0^{2\pi} \frac{1+e^{i6\theta}}{1-2p\cos 2\theta+p^2} d\theta = 2\pi i \cdot \frac{1}{i} \left(\frac{1+p^3}{1-p^2} \right) = 2\pi \left(\frac{1+p^3}{1-p^2} \right).$$

$\therefore$
$$\int_0^{2\pi} \frac{\cos^2 3\theta}{1-2p\cos 2\theta+p^2} d\theta = \frac{1}{2} \text{ real part of } \int_0^{2\pi} \frac{1+e^{i6\theta}}{1-2p\cos 2\theta+p^2} d\theta$$

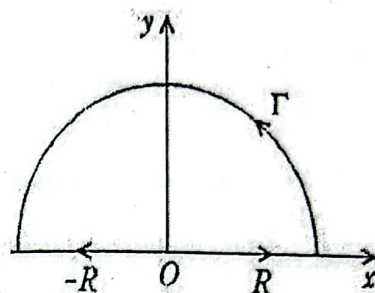
$$= \pi \left(\frac{1+p^3}{1-p^2} \right) = \pi \frac{(1+p+p^2)}{1-p}.$$

Evaluation of the Integral $\int_{-\infty}^{\infty} f(x) dx$

Theorem: If the function $f(z)$ is analytic in the upper half of the z -plane except at a finite number of poles in it, having no poles on the real axis and if further $f(z)$ tends to zero as $|z|$ tends to infinity then by contour integration

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \Sigma R^+$$

where ΣR^+ represents the sum of the residues at the poles in the upper half plane.



Q.

(i) If $a > 0$, prove that $\int_0^{\infty} \frac{dx}{(x^2 + a^2)^2} = \frac{\pi}{4a^3}$.

(ii) Prove that $\int_0^{\infty} \frac{dx}{(1+x^2)^2} = \frac{\pi}{4}$.

Solution: (i) Consider

$$\int_C f(z) dz = \int_C \frac{dz}{(a^2 + z^2)^2}$$

where C is the contour consisting of a large semi-circle Γ of radius R together with real axis from $-R$ to R .

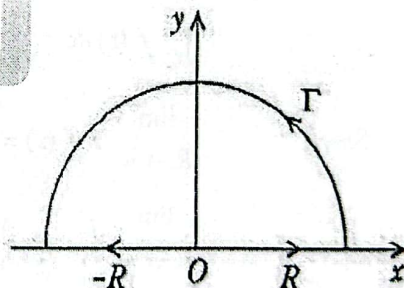
$z = ai, -ai$ are the poles of $f(z)$ of second order. Out of these only $z = ai$ lies inside C .

Residue at $z = ai$ is

$$\phi'(ai) = \left[\left(\frac{d}{dz} \right) (z + ai)^{-2} \right]_{z=ai} = \left[-2(z + ai)^{-3} \right]_{z=ai} = \frac{1}{4a^3 i}$$

By Cauchy's residue theorem, we have

$$\int_C f(z) dz = \int_{-R}^R \frac{dx}{(a^2 + x^2)^2} + \int_{\Gamma} \frac{dz}{(a^2 + z^2)^2} = 2\pi i \Sigma R^+, \quad \dots(1)$$



Now $\lim_{R \rightarrow \infty} z f(z) = \lim_{z \rightarrow \infty} \frac{z}{(a^2 + z^2)^2} = 0,$

$\therefore \lim_{R \rightarrow \infty} \int_{\Gamma} \frac{dz}{(a^2 + z^2)^2} = 0$

and $\lim_{R \rightarrow \infty} \int_{-R}^R \frac{dx}{(a^2 + x^2)^2} = \int_{-\infty}^{\infty} \frac{dx}{(a^2 + x^2)^2}.$

Hence we have from relation (1)

$$\int_{-\infty}^{\infty} \frac{dx}{(a^2 + x^2)^2} = 2 \int_0^{\infty} \frac{dx}{(a^2 + x^2)^2} = 2\pi i \cdot \frac{1}{4a^3 i} = \frac{\pi}{2a^3}$$

or $\int_0^{\infty} \frac{dx}{(a^2 + x^2)^2} = \frac{\pi}{4a^3}.$

(ii) Proceed as in part (i) taking $a = 1$.

Q. Prove that $\int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)^3} dx = \frac{\pi}{8a^3}$, provided that $R(a)$ is +ve. What is the value of this integral when $R(a)$ is negative?

Solution: Consider the integral $\int_C f(z) dz = \int_C \frac{z^2}{(z^2 + a^2)^3} dz,$

where C is the contour as described in Ex. II.

By Cauchy's residue theorem, we have

$$\int_C f(z) dz = \int_{-R}^R \frac{x^2}{(x^2 + a^2)^3} dx + \int_{\Gamma} \frac{z^2}{(z^2 + a^2)^3} dz = 2\pi i \sum R^+ \dots (1)$$

Since $\lim_{R \rightarrow \infty} z f(z) = \lim_{z \rightarrow \infty} z \cdot \frac{z^2}{(z^2 + a^2)^3} = 0$, therefore we have

$$\lim_{R \rightarrow \infty} \int_{\Gamma} \frac{z^2}{(z^2 + a^2)^3} dz = 0.$$

Also $z = \pm ai$ are the poles of $f(z)$ of order three. Out of these only $z = ai$ lies inside C .

Residue at $z = ai$ is $\frac{1}{2!} \left[\frac{d^2}{dz^2} \frac{z^2}{(z + ai)^3} \right]_{z=ai} = \frac{1}{16a^3 i}.$

Now from (1), we have

$$\int_{-\infty}^{\infty} \frac{z^2}{(z^2 + a^2)^3} dz = 2\pi i \cdot \frac{1}{16a^3 i} = \frac{\pi}{8a^3}.$$

If $R(a) < 0$ then $z = -ai$ is the pole inside C .

In this case, $\int_{-\infty}^{\infty} \frac{z^2}{(z^2 + a^2)^3} dz = -\frac{\pi}{8a^3}.$

Aliter: Consider $\int_C \frac{z^2 dz}{(z^2 + a^2)^3} = \int_C f(z) dz$, where C is the same contour as in Ex. 11.

By residue theorem,

$$\int_C f(z) dz = \int_{-R}^R f(x) dx + \int_{\Gamma} f(z) dz = 2\pi i \Sigma R^+.$$

Since $\lim_{z \rightarrow \infty} z f(z) = 0$, we have $\lim_{R \rightarrow \infty} \int_{\Gamma} f(z) dz = 0$.

$\therefore$ Proceeding to the limits, when $R \rightarrow \infty$, we get from (1),

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \Sigma R^+. \quad \dots(2)$$

Now $f(z)$ has poles at $z = \pm ai$ of order three, of which $z = ai$ lies inside C provided $R(a)$ is positive.

Residue of $f(z)$ (at $z = ai$) = coeff. of $\frac{1}{t}$ in $f(t + ai)$.

$$= \text{coeff. of } \frac{1}{t} \text{ in } \frac{(t + ai)^2}{[(t + ai)^2 + a^2]^3}.$$

Now
$$\frac{(t + ai)^2}{(t^2 + 2ait - a^2)^3} = -\frac{1}{8a^3 i t^3} (t^2 + 2ait - a^2) \left(1 + \frac{t}{2ai}\right)^{-3}$$

$$= -\frac{1}{8a^3 i t^3} [t^2 + 2ait - a^2] \left[1 - \frac{3t}{2ai} - \frac{6t^2}{4a^2} + \dots\right]$$

$$\therefore \text{coeff. of } \frac{1}{t} \text{ in } f(t + ai) = -\frac{1}{8a^3 i} \left[1 - \frac{6ai}{2ai} + \frac{6a^2}{4a^2}\right] = \frac{1}{16a^3 i}.$$

$$\therefore \text{Residue of } f(z) \text{ (at } z = ai) = \frac{1}{16a^3 i}.$$

$$\therefore \text{From (2), } \int_{-\infty}^{\infty} f(x) dx = 2\pi i \cdot \frac{1}{16a^3 i} = \frac{\pi}{8a^3}. \quad \dots(3)$$

When $R(a)$ is -ve, pole within C is at $z = -ai$.

$$\therefore \text{In this case } \int_{-\infty}^{\infty} f(x) dx = -\frac{\pi}{8a^3}. \quad [\text{Replacing } -a \text{ by } a \text{ in (3)}]$$

Q. Evaluate $\int_0^{\infty} \frac{dx}{x^4 + a^4}, a > 0$.

Solution: Consider the integral $\int_C f(z) dz = \int_C \frac{dz}{z^4 + a^4}$,

where C is the contour as described in Ex. 11.

By Cauchy's residue theorem, we have

$$\int_C f(z) dz = \int_{-R}^R \frac{dx}{x^4 + a^4} + \int_{\Gamma} \frac{dz}{z^4 + a^4} = 2\pi i \Sigma R^+$$

Since $\lim_{R \rightarrow \infty} z f(z) = 0$ therefore $\lim_{R \rightarrow \infty} \int_{\Gamma} \frac{dz}{z^4 + a^4} = 0$

so that $\lim_{R \rightarrow \infty} \int_{-R}^R \frac{dx}{x^4 + a^4} = 2\pi i \Sigma R^+$

or
$$\int_{-\infty}^{\infty} \frac{dx}{x^4 + a^4} = 2\pi i \Sigma R^+ \quad \dots(1)$$

Poles of $f(z)$ are given by $z^4 + a^4 = 0$

or $z = a e^{(2n+1)\pi i/4}$, where $n = 0, 1, 2, 3$.

$\therefore z = a e^{i\pi/4}, a e^{3i\pi/4}, a e^{5i\pi/4}, a e^{7i\pi/4}$ are the simple poles of $f(z)$.

Out of these only $z = a e^{i\pi/4}, a e^{3i\pi/4}$ lie inside C .

If α denotes any of these poles then residue at $z = \alpha$ is

$$\begin{aligned} \lim_{z \rightarrow \alpha} (z - \alpha) f(z) &= \lim_{z \rightarrow \alpha} \frac{z - \alpha}{z^4 + a^4} \quad \left[\text{Form } \frac{0}{0} \right] \\ &= \lim_{z \rightarrow \alpha} \frac{1}{4z^3} = \frac{1}{4\alpha^3} = \frac{\alpha}{4\alpha^4} = -\frac{\alpha}{4a^4} \quad [\because \alpha^4 = -a^4] \end{aligned}$$

Now sum of the residues at poles inside C

$$= -\frac{1}{4a^4} \{a e^{i\pi/4} + a e^{3i\pi/4}\} = -\frac{1}{4a^3} \cdot \frac{2i}{\sqrt{2}} = -\frac{i\sqrt{2}}{4a^3}$$

Hence from (1), we have

$$\int_{-\infty}^{\infty} \frac{dx}{x^4 + a^4} = 2\pi i \left(-\frac{i\sqrt{2}}{4a^3} \right) = \frac{\pi\sqrt{2}}{2a^3}$$

or
$$\int_0^{\infty} \frac{dx}{x^4 + a^4} = \frac{\pi\sqrt{2}}{4a^3}$$

Jordan's Inequality. $\frac{2\theta}{\pi} \leq \sin \theta \leq \theta$, where $0 \leq \theta \leq \frac{\pi}{2}$.

Let $y = \cos x$. We know that as θ increases from 0 to $\pi/2$, $\cos \theta$ decreases from 1 to 0. Consequently the mean ordinate of the graph $y = \cos x$ also decreases steadily over the range $0 \leq x \leq \theta$.

The mean ordinate is $\frac{1}{\theta} \int_0^\theta \cos \theta \, d\theta = \frac{\sin \theta}{\theta}$.

Hence when $0 \leq \theta \leq \frac{\pi}{2}$, $\frac{2}{\pi} \leq \frac{\sin \theta}{\theta} \leq 1$ or $\frac{2\theta}{\pi} \leq \sin \theta \leq \theta$.

This is known as Jordan's inequality.

Jordan's Lemma

If $f(z)$ tends to zero uniformly as $z \rightarrow \infty$ and $f(z)$ is meromorphic in the upper half plane then

$$\lim_{R \rightarrow \infty} \int_{\Gamma} e^{imz} f(z) \, dz = 0, \quad (m > 0)$$

where Γ denotes the semi-circle $|z| = R$, $I(z) > 0$.

Proof: Here we assume that R is large enough so as to include within it all the singularities of $f(z)$ and $f(z)$ has no singularity on Γ .

Since $\lim_{R \rightarrow \infty} f(z) = 0$ therefore for $\epsilon > 0$ there exists $R_0 > 0$ such that $|f(z)| < \epsilon$ when $|z| = R \leq R_0$.

Now let Γ denote any semi-circle with radius $R \geq R_0$. Putting $z = Re^{i\theta}$, we get

$$\begin{aligned}\int_{\Gamma} e^{imz} f(z) dz &= \int_0^{\pi} e^{imRe^{i\theta}} f(Re^{i\theta}) R i e^{i\theta} d\theta \\ &= \int_0^{\pi} e^{imR \cos \theta} e^{-mR \sin \theta} f(Re^{i\theta}) i R e^{i\theta} d\theta. \\ \left| \int_{\Gamma} e^{imz} f(z) dz \right| &\leq \int_0^{\pi} |e^{imR \cos \theta}| |e^{-mR \sin \theta}| |f(Re^{i\theta})| |R i e^{i\theta}| d\theta \\ &< \int_0^{\pi} e^{-mR \sin \theta} \epsilon R d\theta \quad [\because |f(z)| = |f(Re^{i\theta})| < \epsilon] \\ &= 2\epsilon R \int_0^{\pi/2} e^{-mR \sin \theta} d\theta \\ &\leq 2\epsilon R \int_0^{\pi/2} e^{-2mR \theta/\pi} d\theta, \text{ by Jordan's inequality} \\ &= \frac{2\epsilon R (1 - e^{-mR})}{2mR/\pi} = \frac{\epsilon\pi}{m} (1 - e^{-mR}) < \frac{\epsilon\pi}{m}.\end{aligned}$$

Hence $\lim_{R \rightarrow \infty} \int_{\Gamma} e^{imz} f(z) dz = 0$.

Evaluation of the Integrals of the form

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} \sin mx \, dx; \int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} \cos mx \, dx, m > 0, \text{ where}$$

- (i) $P(x), Q(x)$ are polynomials, (ii) $\deg Q(x) > \deg P(x)$
- (iii) $Q(x) = 0$ has no real roots.

Under the above mentioned conditions the given integrals are convergent. Consider

$$\int_C e^{imz} f(z) dz = \int_C e^{imz} \frac{P(z)}{Q(z)} dz,$$

where C is the contour consisting of a semi-circle Γ of radius R so large as to include all the poles of the integrand in the upper half plane and part of the real axis from $-R$ to R .

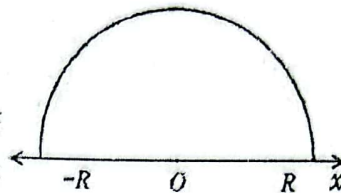
Q. Prove that $\int_0^\infty \frac{\cos mx}{a^2 + x^2} dx = \frac{\pi}{2a} e^{-ma}$, $m \geq 0$.

Deduce that $\int_0^\infty \frac{x \sin mx}{x^2 + a^2} dx = \frac{\pi}{2} e^{-ma}$.

Solution: Consider the integral

$$\int_C \frac{e^{imz}}{a^2 + z^2} dz = \int_C f(z) dz,$$

where C is the contour consisting of a large semi-circle Γ of radius R containing all the poles of the integrand in the upper half plane and the part of real axis from $-R$ to R .



By Cauchy's residue theorem, we have

$$\int_C f(z) dz = \int_{-R}^R \frac{e^{imx}}{a^2 + x^2} dx + \int_\Gamma \frac{e^{imz}}{a^2 + z^2} dz = 2\pi i \Sigma R^+.$$

Since $\lim_{z \rightarrow \infty} \frac{1}{z^2 + a^2} = 0$ therefore we have

$$\lim_{R \rightarrow \infty} \int_\Gamma f(z) dz = 0, \text{ by Jordan's lemma.}$$

$$\therefore \lim_{R \rightarrow \infty} \int_{-R}^R \frac{e^{imx}}{a^2 + x^2} dx = 2\pi i \Sigma R^+$$

$$\text{or} \quad \int_{-\infty}^{\infty} \frac{e^{imx}}{a^2 + x^2} dx = 2\pi i \Sigma R^+ \quad \dots(1)$$

$z = \pm ai$ are the simple poles of $f(z)$. The pole $z = ai$ lies inside C .

Residue at $z = ai$ is

$$\lim_{z \rightarrow ai} (z - ai) f(z) = \lim_{z \rightarrow ai} \frac{e^{imz}}{z^2 + a^2} (z - ai) = \frac{e^{-ma}}{2ia}$$

From (1), we have

$$\int_{-\infty}^{\infty} \frac{e^{imx}}{x^2 + a^2} dx = \frac{2\pi i e^{-ma}}{2ia} = \frac{\pi}{a} e^{-ma}.$$

Equating real parts on both sides, we get

$$\int_{-\infty}^{\infty} \frac{\cos mx}{x^2 + a^2} dx = \frac{\pi}{a} e^{-ma} \quad \text{or} \quad \int_0^\infty \frac{\cos mx}{x^2 + a^2} dx = \frac{\pi}{2a} e^{-ma}.$$

Differentiating both sides w.r.t. m , we get

$$\int_0^\infty \frac{x \sin mx}{a^2 + x^2} dx = \frac{\pi}{2a} (-a) e^{-ma} \quad \text{or} \quad \int_0^\infty \frac{x \sin mx}{a^2 + x^2} dx = \frac{\pi}{2} e^{-ma}.$$

Q. Evaluate $\int_0^{\infty} \frac{\cos ax}{(x^2 + b^2)^2} dx$, $a > 0, b > 0$.

Solution: Consider $\int_C \frac{e^{iaz}}{(z^2 + b^2)^2} dz = \int_C f(z) dz$,

where C is the same contour as described in Ex. 15.

By Cauchy's residue theorem, we have

$$\int_C f(z) dz = \int_{-R}^R \frac{e^{iax}}{(x^2 + b^2)^2} dx + \int_{\Gamma} \frac{e^{iaz}}{(z^2 + b^2)^2} dz = 2\pi i \Sigma R^+.$$

By Jordan's lemma, we have

$$\lim_{R \rightarrow \infty} \int_{\Gamma} \frac{e^{iaz}}{(z^2 + b^2)^2} dz = 0.$$

$$\therefore \lim_{R \rightarrow \infty} \int_{-R}^R \frac{e^{iax}}{(x^2 + b^2)^2} dx = 2\pi i \Sigma R^+$$

$$\text{or} \quad \int_{-\infty}^{\infty} \frac{e^{iax}}{(x^2 + b^2)^2} dx = 2\pi i \Sigma R^+ \quad \dots(1)$$

$z = \pm ib$ are the double poles of $f(z)$. Out of these only $z = ib$ lies in the upper half plane.

Residue at $z = ib$ is

$$\begin{aligned} \phi'(ib) &= \left[D \frac{e^{iaz}}{(z + ib)^2} \right]_{z=ib} \\ &= \left[\frac{iae^{iaz}(z + ib)^2 - 2e^{iaz}(z + ib)}{(z + ib)^4} \right]_{z=ib} \\ &= \frac{e^{-ab} [ia(2ib) - 2]}{(2ib)^3} = \frac{e^{-ab}}{4b^3 i} (ab + 1). \end{aligned}$$

Hence from (1), we have

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{(x^2 + b^2)^2} dx = 2\pi i \cdot \frac{e^{-ab}}{4ib^3} (ab + 1)$$

$$\text{or} \quad \int_{-\infty}^{\infty} \frac{\cos ax + i \sin ax}{(x^2 + b^2)^2} dx = \frac{\pi e^{-ab}}{2b^3} (ab + 1).$$

Equating real parts on both sides, we get

$$\int_{-\infty}^{\infty} \frac{\cos ax}{(x^2 + b^2)^2} dx = \frac{\pi (ab + 1)}{2b^3 e^{ab}}$$

$$\text{or} \quad \int_0^{\infty} \frac{\cos ax}{(x^2 + b^2)^2} dx = \frac{\pi (1 + ab)}{4b^3 e^{ab}}.$$

Q.

Prove that

$$\int_{-\infty}^{\infty} \frac{\sin x}{(1-x+x^2)^2} dx = \frac{2\pi(\sqrt{3}+2)}{3\sqrt{3}} e^{-\sqrt{3}/2} \sin \frac{1}{2}.$$

Solution: Let $\int_C f(z) dz = \int_C \frac{e^{iz}}{(1-z+z^2)^2} dz$, where C is the same contour as in

Ex. 15.

By residue theorem, we have

$$\int_C f(z) dz = \int_{-R}^R f(x) dx + \int_{\Gamma} f(z) dz = 2\pi i \Sigma R^+.$$

By Jordan's lemma, we have

$$\lim_{R \rightarrow \infty} \int_{\Gamma} f(z) dz = 0.$$

$$\therefore \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx = 2\pi i \Sigma R^+$$

$$\text{or} \quad \int_{-\infty}^{\infty} f(x) dx = 2\pi i \Sigma R^+. \quad \dots(1)$$

$z = (1 \pm i\sqrt{3})/2$ are the poles of $f(z)$ of second order. The only pole which lies within C is $(1 + i\sqrt{3})/2 = \alpha$, say.

Putting $z = \alpha + t$ in $f(z)$, we get

$$\begin{aligned} f(\alpha + t) &= \frac{e^{i(\alpha+t)}}{\{t^2 + (2\alpha-1)t + \alpha^2 - \alpha + 1\}^2} \\ &= \frac{e^{i(\alpha+t)}}{\{t^2 + (2\alpha-1)t\}^2}, \text{ since } \alpha^2 - \alpha + 1 = 0 \\ &= \frac{e^{i(\alpha+t)}}{(2\alpha-1)^2 t^2} \left[1 + \frac{t}{2\alpha-1}\right]^{-2} \\ &= \frac{e^{i\alpha}}{(2\alpha-1)^2 t^2} (1 + it + \dots) \left(1 - \frac{2t}{2\alpha-1} + \dots\right). \end{aligned}$$

Residue at $z = \alpha$ is the coefficient of $(1/t)$ in the expansion of

$$\begin{aligned} f(\alpha + t) &= e^{i\alpha} \left[\frac{i}{(2\alpha-1)^2} - \frac{2}{(2\alpha-1)^3} \right] \\ &= e^{i\alpha} \left[\frac{i(2\alpha-1) - 2}{(2\alpha-1)^3} \right] \\ &= e^{i(1+i\sqrt{3})/2} \frac{[i(1+i\sqrt{3}-1) - 2]}{(i\sqrt{3})^3} \end{aligned}$$

$$= \frac{e^{-\sqrt{3}/2} e^{i/2} (\sqrt{3} + 2)}{i 3 \sqrt{3}}.$$

Hence from (1), we have

$$\int_{-\infty}^{\infty} \frac{e^{ix}}{(x^2 - x + 1)^2} dx = \frac{2\pi e^{-\sqrt{3}/2}}{3 \sqrt{3}} (\sqrt{3} + 2) e^{i/2}.$$

Equating imaginary parts on both sides, we get

$$\int_{-\infty}^{\infty} \frac{\sin x}{(x^2 - x + 1)^2} dx = \frac{2\pi e^{-\sqrt{3}/2}}{3 \sqrt{3}} (\sqrt{3} + 2) \sin \frac{1}{2}.$$

Q. Prove that $\int_0^{\infty} \frac{\cos x^2 + \sin x^2 - 1}{x^2} dx = 0$.

Solution: Let $\int_C f(z) dz = \int_C \frac{e^{iz^2} - 1}{z^2} dz$, where C is the same contour as in Example 15.

Since $f(z)$ has no poles in the upper half plane therefore by Cauchy's residue theorem, we have

$$\int_C f(z) dz = \int_{-R}^R f(x) dx + \int_{\Gamma} f(z) dz = 0. \quad \dots(1)$$

We have
$$\begin{aligned} \int_{\Gamma} f(z) dz &= \int_0^{\pi} \frac{\exp(i R^2 e^{i2\theta}) - 1}{R^2 e^{i2\theta}} R i e^{i\theta} d\theta, \text{ putting } z = R e^{i\theta} \\ &= \frac{i}{R} \int_0^{\pi} e^{-i\theta} [\exp\{i R^2 (\cos 2\theta + i \sin 2\theta)\} - 1] d\theta \\ &= \int_0^{\pi} \frac{i}{R} e^{-i\theta} [\exp(-R^2 \sin 2\theta) \exp(i R^2 \cos 2\theta) - 1] d\theta. \end{aligned}$$

Now
$$\begin{aligned} \left| \int_{\Gamma} f(z) dz \right| &\leq \frac{1}{R} \int_0^{\pi} |i e^{-i\theta}| [\exp(-R^2 \sin 2\theta) |\exp(i R^2 \cos 2\theta)| + |-1|] d\theta \\ &\leq \frac{1}{R} \int_0^{\pi} [\exp(-R^2 \sin 2\theta) + 1] d\theta, \text{ which tends to zero as } R \rightarrow \infty. \end{aligned}$$

$\therefore \int_{\Gamma} f(z) dz = 0$ when $R \rightarrow \infty$.

Thus when $R \rightarrow \infty$, we have from (1)

$$\int_{-\infty}^{\infty} f(x) dx = 0 \quad \text{or} \quad \int_{-\infty}^{\infty} \frac{e^{ix^2} - 1}{x^2} dx = 0$$

or
$$2 \int_0^{\infty} \frac{e^{ix^2} - 1}{x^2} dx = 0,$$

Equating real and imaginary parts on both sides, we get

$$\int_0^{\infty} \frac{\cos x^2 - 1}{x^2} dx = 0 \quad \text{and} \quad \int_0^{\infty} \frac{\sin x^2}{x^2} dx = 0.$$

Adding these relations, we get

$$\int_0^{\infty} \frac{\cos x^2 + \sin x^2 - 1}{x^2} dx = 0.$$

Q. Prove by contour integration $\int_0^{\infty} \frac{\log(1+x^2)}{1+x^2} dx = \pi \log 2$.

Solution: Consider

$$\int_C \frac{\log(i+z)}{1+z^2} dz = \int_C f(z) dz, \text{ where } C \text{ is the contour of Ex. 15.}$$

By residue theorem, we have

$$\int_C f(z) dz = \int_{-R}^R f(x) dx + \int_{\Gamma} f(z) dz = 2\pi i \sum R^+. \quad \dots(1)$$

Now

$$\begin{aligned} \lim_{z \rightarrow \infty} z f(z) &= \lim_{z \rightarrow \infty} \frac{z \log(i+z)}{(i+z)(z-i)} \\ &= \lim_{z \rightarrow \infty} \frac{z}{z-i} \cdot \lim_{z \rightarrow \infty} \frac{\log(i+z)}{i+z} = 0. \end{aligned}$$

$\therefore$ when $R \rightarrow \infty$, we have from (1)

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum R^+ \quad \dots(2)$$

$z = \pm i$ are the simple poles of $f(z)$ and only $z = i$ lies inside C .

Residue at $z = i$ is

$$\lim_{z \rightarrow i} (z-i) f(z) = \lim_{z \rightarrow i} \frac{\log(i+z)}{(z+i)} = \frac{\log 2i}{2i} = \frac{\log 2 + i(\pi/2)}{2i}.$$

Hence from (2), we have

$$\int_{-\infty}^{\infty} \frac{\log(i+x)}{1+x^2} dx = \pi \{\log 2 + i(\pi/2)\}.$$

Equating real parts on both sides, we get

$$\int_{-\infty}^{\infty} \frac{\frac{1}{2} \log(x^2+1)}{x^2+1} dx = \pi \log 2 \quad \text{or} \quad \int_0^{\infty} \frac{\log(1+x^2)}{1+x^2} dx = \pi \log 2.$$

Poles Lie on the Real Axis

We shall now discuss the case when the integrand has poles on the real axis as well as within the semi-circle.

Theorem: Let $f(z) = \frac{P(z)}{Q(z)}$, where $P(z)$ and $Q(z)$ are polynomials such that $Q(z)$ has only non-repeated real roots, that is $f(z)$ has only simple poles on the real axis. Let $m > 0$ and let the degree of $Q(z)$ exceeds that of $P(z)$, then

$$P \int_{-\infty}^{\infty} e^{imx} f(x) dx = 2\pi i \sum_{k=1}^p \text{Res}(a_k) + \pi i \sum_{k=1}^q \text{Res}(b_k),$$

where $a_1, a_2, \dots, a_p$ are the zeros of $Q(z)$ in the region $\text{Im } z > 0$ and $b_1, b_2, \dots, b_q$ are its zeros in the real axis, and $\text{Res}(\alpha)$ denotes the residue of $e^{imz} f(z)$ at α .

Q. If $m > 0$, show that $P \int_{-\infty}^{\infty} \frac{\cos mx}{x-b} dx = -\sin mb$.

Solution: Referring to the above theorem, let $f(z) = \frac{1}{z-b}$. Here $f(z)$ has simple real pole at $z = b$.

$$\therefore \text{Res}_{z=b} \frac{e^{imz}}{z-b} = \lim_{z \rightarrow b} (z-b) \cdot \frac{e^{imz}}{(z-b)} = e^{imb}.$$

Hence
$$P \int_{-\infty}^{\infty} \frac{e^{imz}}{z-b} dz = \pi i e^{imb}.$$

Equating real parts on both sides, we get

$$P \int_{-\infty}^{\infty} \frac{\cos mx}{x-b} dx = -\pi \sin mb.$$

Q. If $m > 0$, prove that $P \int_0^{\infty} \frac{\sin mx}{x} dx = \frac{\pi}{2}$.

Solution: Referring to the above theorem, let $f(z) = \frac{1}{z}$. Here $f(z)$ has simple real pole at $z = 0$.

$$\therefore \text{Res}_{z=0} \frac{e^{imz}}{z} = \lim_{z \rightarrow 0} (z-0) \frac{e^{imz}}{z} = 1.$$

Thus
$$P \int_{-\infty}^{\infty} \frac{e^{imx}}{x} dx = \pi i \times \text{Residue of } \frac{e^{imz}}{z} \text{ at } z = 0$$

$$= \pi i \times 1 = \pi i.$$

Equating real and imaginary parts on both sides, we get

$$P \int_{-\infty}^{\infty} \frac{\cos mx}{x} dx = 0 \quad \dots(1)$$

and
$$P \int_{-\infty}^{\infty} \frac{\sin mx}{x} dx = \pi. \quad \dots(2)$$

Note: The principal part 'P' has been dropped in (2), since $\lim_{x \rightarrow 0} \frac{\sin mx}{x} = m$,

whereas in the first integral the integrand becomes unbounded at the origin.

Hence from (2), we get the required result.

Indenting Method: We can avoid the poles which lie on the real axis by drawing semi-circles of small radii about these poles as centres. This method is known as 'indenting at a point'.

Q. (i) Evaluate $\int_0^\infty \frac{\sin mx}{x} dx, m > 0$. (Gorakhpur 2007, 10, 13; Kumaun 07)

(ii) By integrating e^{iz} / z around a suitable contour, prove that

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}. \quad (\text{Rohilkhand 2011; Kumaun 14})$$

Solution: (i) Consider the integral $\int_C f(z) dz = \int_C \frac{e^{imz}}{z} dz$

where C is the contour consisting of (1) the upper half of the circle $|z| = R$

(2) real axis from r to R where r is small and R is large

(3) real axis from $-R$ to $-r$

(4) upper half of the circle $\gamma, |z| = r$.

Obviously the function $f(z)$ has no singularity inside C , therefore by Cauchy's residue theorem, we have

$$\begin{aligned} \int_C f(z) dz &= \int_r^R f(x) dx + \int_\Gamma f(z) dz + \int_{-R}^{-r} f(x) dx + \int_\gamma f(z) dz \\ &= 0. \end{aligned} \quad \dots(1)$$

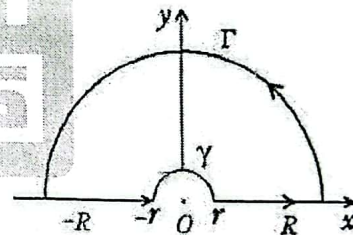
By Jordan's lemma, we have

$$\lim_{R \rightarrow \infty} \int_\Gamma f(z) dz = 0.$$

Again

$$\lim_{z \rightarrow 0} z f(z) = 1 \text{ therefore}$$

$$\lim_{r \rightarrow 0} \int_\gamma f(z) dz = i(0 - \pi) = -i\pi.$$



Thus when $r \rightarrow 0, R \rightarrow \infty$, we get from (1)

$$\int_0^\infty f(x) dx + \int_{-\infty}^0 f(x) dx - i\pi = 0$$

or $\int_{-\infty}^\infty f(x) dx = i\pi$ or $\int_{-\infty}^\infty \frac{e^{imx}}{x} dx = i\pi.$

Equating imaginary parts on both sides, we get

$$\int_{-\infty}^\infty \frac{\sin mx}{x} dx = \pi \quad \text{or} \quad \int_0^\infty \frac{\sin mx}{x} dx = \frac{\pi}{2}.$$

(ii) Proceed as in part (i) taking $m = 1$.

Q. Prove that if $a > 0$

(i) $P \int_{-\infty}^\infty \frac{\cos x}{a^2 - x^2} dx = \frac{\pi \sin a}{a}$

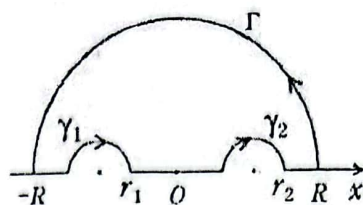
(ii) $P \int_{-\infty}^\infty \frac{\sin x}{a^2 - x^2} dx = 0.$

(Gorakhpur 2008, 09)

Solution: Consider the integral

$$\int_C f(z) dz = \int_C \frac{e^{iz}}{a^2 - z^2} dz,$$

where C is the contour consisting of the large semi-circle Γ of radius R , indented at $z = a$ and $z = -a$. γ_1 and γ_2 are the small semi-circles with radii r_1 and r_2 and centres at $z = -a$ and $z = a$ respectively.



The function $f(z)$ is regular within and on the contour C therefore by Cauchy's residue theorem, we have

$$\begin{aligned} \int_C f(z) dz &= \int_{\Gamma} f(z) dz + \int_{-R}^{-(a+r_1)} f(x) dx + \int_{\gamma_1} f(z) dz \\ &+ \int_{-(a-r_1)}^a f(x) dx + \int_{\gamma_2} f(z) dz + \int_{a+r_2}^R f(x) dx = 0. \end{aligned} \quad \dots(1)$$

Since $1/(a^2 - z^2) \rightarrow 0$ as $z \rightarrow \infty$, therefore by Jordan's lemma, we have

$$\lim_{R \rightarrow \infty} \int_{\Gamma} f(z) dz = 0.$$

Also

$$\lim_{z \rightarrow -a} (a+z) f(z) = \lim_{z \rightarrow -a} \frac{e^{iz}}{a-z} = \frac{e^{-ia}}{2a},$$

therefore

$$\lim_{r_1 \rightarrow 0} \int_{\gamma_1} f(z) dz = i \frac{e^{-ia}}{2a} (0 - \pi) = -i \frac{\pi}{2a} e^{-ia}.$$

Similarly

$$\lim_{r_2 \rightarrow 0} \int_{\gamma_2} f(z) dz = i \frac{\pi}{2a} e^{ia}.$$

Hence when $R \rightarrow \infty, r_1 \rightarrow 0, r_2 \rightarrow 0$, we have from (1)

$$\int_{-\infty}^{-a} f(x) dx + \left(-i \frac{\pi}{2a} e^{-ia}\right) + \int_{-a}^a f(x) dx + \frac{i\pi}{2a} e^{ia} + \int_a^{\infty} f(x) dx = 0$$

or

$$P \int_{-\infty}^{\infty} f(x) dx = -i \frac{\pi}{2a} (e^{ia} - e^{-ia})$$

or

$$P \int_{-\infty}^{\infty} \frac{e^{ix}}{a^2 - x^2} dx = \frac{\pi}{a} \sin a.$$

Equating real and imaginary parts on both sides, we get

$$P \int_{-\infty}^{\infty} \frac{\cos x}{a^2 - x^2} dx = \frac{\pi}{a} \sin a, \quad P \int_{-\infty}^{\infty} \frac{\sin x}{a^2 - x^2} dx = 0.$$

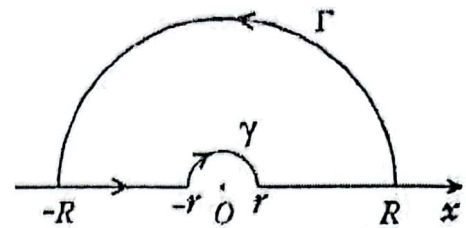
Q.

$$\text{Prove that } \int_0^{\infty} \frac{\sin x}{x(x^2 + a^2)} dx = \frac{\pi}{2a^2} (1 - e^{-a}), a > 0.$$

Solution: Consider the integral $\int_C f(z) dz = \int_C \frac{e^{iz}}{z(z^2 + a^2)} dz,$

where C is the contour consisting of

- (i) a large semi-circle $\Gamma, |z| = R$ in the upper half plane
- (ii) real axis from r to R
- (iii) real axis from $-R$ to $-r$
- (iv) a small semi-circle $\gamma, |z| = r$.



$z = 0, \pm ia$ are the simple poles of $f(z)$. Out of these only $z = ia$ lies within C .

Residue at $z = ai$ is

$$\lim_{z \rightarrow ai} (z - ai) f(z) = \lim_{z \rightarrow ai} \frac{e^{iz}}{z(z + ai)} = \frac{e^{-a}}{-2a^2}.$$

By residue theorem, we have

$$\begin{aligned} \int_C f(z) dz &= \int_r^R f(x) dx + \int_\Gamma f(z) dz + \int_{-R}^{-r} f(x) dx + \int_\gamma f(z) dz \\ &= 2\pi i \Sigma R^+. \end{aligned} \quad \dots(1)$$

By Jordan's lemma, we have $\lim_{R \rightarrow \infty} \int_\Gamma f(z) dz = 0$.

Also $\lim_{z \rightarrow 0} z f(z) = \lim_{z \rightarrow 0} \frac{e^{iz}}{z^2 + a^2} = \frac{1}{a^2},$

$\therefore \lim_{r \rightarrow 0} \int_\gamma f(z) dz = i \cdot \frac{1}{a^2} (0 - \pi) = -\frac{\pi i}{a^2}.$

$\therefore$ when $r \rightarrow 0$ and $R \rightarrow \infty$, we have from (1)

$$\int_0^\infty f(x) dx + \int_{-\infty}^0 f(x) dx - \frac{i\pi}{a^2} = 2\pi i \left(-\frac{e^{-a}}{2a^2} \right)$$

or $\int_{-\infty}^\infty f(x) dx = \frac{i\pi}{a^2} (1 - e^{-a}).$

Equating imaginary parts on both sides, we get

$$\int_{-\infty}^\infty \frac{\sin x}{x(x^2 + a^2)} dx = \frac{\pi}{a^2} (1 - e^{-a}).$$